

# RUNHAN HUANG

Harvard University

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## EDUCATION

**Harvard University, John A. Paulson School of Engineering and Applied Sciences**

Aug 2026 - Now

*Ph.D. in Computer Science*

Cambridge, MA

**Tsinghua University, Yao Class**

September 2022 - June 2026

*B.Eng. in Computer Science (GPA: 3.9 / 4.0)*

Beijing

## EXPERIENCE

**MARS Lab, IIIS & Shanghai Qizhi Institute**

June 2024 – March 2025

*Research Assistant*

Beijing

- Advised by **Prof. Hang Zhao**, working on *Robot Locomotion* and *Robot Manipulation*

**Du Lab, Kempner Institute, Harvard University**

March 2025 – September 2025

*Visiting Researcher*

MA, USA

- Advised by **Prof. Yilun Du**, working on *Diffusion Planning*

## PUBLICATIONS

- [P8] Zhangchen Ye\*, Enxuan Ruan\*, Yifei Bao\*, **Runhan Huang**, Jiankun Yang, Jiakang Jin, Yixiao Huo, Pengyuan Wang, Yinan Han, Huaxing Huang, Wenhao Cui, Yiming Li†, Xiaoyu Tian†. “*SkillX: Unified Multi-Skill Policy Learning for Humanoid Soccer*,” *CoRL*, 2026.
- [P7] Zunzhe Zhang\*, **Runhan Huang\***, Yicheng Liu, Shaoting Zhu, Linzhan Mou, Hang Zhao. “*Generative Control as Optimization: Time-Unconditional Flow Matching for Adaptive and Robust Robotic Control*,” *NeurIPS*, 2026.
- [P6] Bohan Lyu\*, Yucheng Yang\*, Siqiao Huang\*, Jiaru Zhang\*, Qixin Xu\*, Xinghan Li\*, Xinyang Han\*, Yicheng Zhang\*, Huaqing Zhang\*, **Runhan Huang**, et al.. “*MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI*,” *NeurIPS (Oral)*, 2026.
- [P5] Shaoting Zhu\*, Baijun Ye\*, Jiaxuan Wang†, Jiakang Chen†, Ziwen Zhuang, Linzhan Mou, **Runhan Huang**, Hang Zhao. “*TTT-Parkour: Rapid Test-Time Training for Perceptive Robot Parkour*,” *CoRL*, 2026.
- [P4] **Runhan Huang**, Haldun Balim, Heng Yang, Yilun Du. “*Flexible Locomotion Learning with Diffusion Model Predictive Control*,” *ICRA*, 2026.
- [P3] **Runhan Huang\***, Shaoting Zhu\*, Yilun Du, Hang Zhao. “*MoELoco: Mixture of Experts for Multitask Locomotion*,” *IROS*, 2025.
- [P2] Shaoting Zhu\*, Linzhan Mou\*, Derun Li, Baijun Ye, **Runhan Huang**, Hang Zhao. “*VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion*,” *RA-L*, 2025.
- [P1] Shaoting Zhu, **Runhan Huang**, Linzhan Mou, Hang Zhao. “*Robust Robot Walker: Learning Agile Locomotion over Tiny Traps*,” *ICRA*, 2025.

## RESEARCH EXPERIENCE

### Generative Control as Optimization | *ICLR Workshops 2026*

- Reinterpret flow-matching inference from fixed-schedule integration to iterative optimization by learning a time-unconditional, stationary velocity field
- Introduce an adaptive inference mechanism that dynamically allocates computation for contact-rich bottlenecks, alongside a training-free OOD anomaly detector using the vector field norm
- Scale the framework to  $\pi_0$ -series VLA models and demonstrate superior sample efficiency and real-world robustness on a Galaxea R1 Lite humanoid robot for high-precision tasks

### TTT-Parkour: Rapid Test-Time Training for Perceptive Robot Parkour | *CoRL 2026*

- Develop a real-to-sim-to-real framework that rapidly adapts humanoid locomotion policies to unseen and highly challenging terrain geometries through test-time training

- Build an efficient RGB-D geometry reconstruction pipeline and fine-tune a pretrained locomotion policy on high-fidelity terrain meshes in under 10 minutes
- Demonstrate robust zero-shot sim-to-real transfer across wedges, stakes, boxes, trapezoids, narrow beams, and mixed-terrain parkour courses

#### **Flexible Locomotion Learning with Diffusion Model Predictive Control | ICRA 2026**

- Reinterpret diffusion models as generative priors for MPC, using reward updates, constraint projection, and candidate ranking
- Introduce a reward-weighted denoising method to finetune the diffusion planner, improving locomotion performance
- Develop real-time diffusion planning techniques such as asynchronous execution and early caching; demonstrate zero-shot transfer on a Unitree Go2 quadruped

#### **MoELoco: Mixture of Experts for Multitask Locomotion | IROS 2025**

- Develop a multitask locomotion framework using a mixture-of-experts (MoE) architecture.
- Enhance sample efficiency and skill compositionality for reinforcement learning across multiple locomotion tasks.
- Project leader; contributed to model design, training infrastructure, evaluation and real-world deployment.

#### **VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion | RA-L 2025**

- Build a digital-twin simulation pipeline using 3D Gaussian Splatting for photo-realistic rendering.
- Propose an RGB-based sim-to-real transfer policy framework for navigation and locomotion.
- Participate in low-level locomotion training and real world experiment

#### **Robust Robot Walker: Learning Agile Locomotion over Tiny Traps | ICRA 2025**

- Design a proprioception-only locomotion policy that generalizes to terrains with small obstacles ("tiny traps").
- Introduce a two-stage reinforcement learning training framework and a novel benchmark for tiny-trap traversal.
- Co-lead benchmark design, policy training, evaluation and real-robot deployment.

## **PROJECTS**

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#### **Implicitly Exploit the Modularity in Multimodal Large Language Models | Multimodal Machine Learning Course Project**

- Propose a new gradient update strategy for visual instruction tuning
- Train a LVLM based on phi-2 model, achieving good performance

#### **Humanoid Perceptive Locomotion | Intelligent System and Robotics Course Project**

- Construct a two-stage teacher-student locomotion learning paradigm with PPO and DAgger, enabling a humanoid robot to traverse challenging terrain
- Demonstrate robust stair, slope, and gap traversal in simulation using only depth image and proprioception

#### **Multi-agent Navigation with Vision Language Model | Practical Research Training Project**

- Build a city-scale multi-agent navigation dataset combining Google Street View, autonomous driving data, and 3D Gaussian Splatting reconstructions, enriched with landmark, orientation, and action annotations to directly supervise spatial reasoning skills.
- Design a dual benchmarking protocol spanning static VQA-style prediction and 3DGS-based simulation, providing a comprehensive assessment of perception, planning, and coordination
- Benchmark the state-of-the-art VLMs, showing that current models fall significantly short, motivating new research on embodied multi-agent intelligence

## **SKILLS**

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**Programming:** C, C++, Python, Pytorch, ROS, ROS2, Typescript, Scala

**Robotics related software/hardware experience:** isaacgym, Unitree A1, Unitree Go2, Unitree G1, Galaxea R1 Lite Mobile Manipulator, PiperX Robot Arm, Franka

**Key Course Taken:** C/C++ Programming (A+), Calculus (A), Linear Algebra (A), General Physics (A+), Algebra and Computation (A+), Algorithm Design, Probability and Statistic, Machine Learning, Deep Learning, Multimodal Machine Learning (A+), Computer Vision (A), Theoretical Computer Science, Reinforcement Learning, Autonomous Driving (A+), Advanced Computer Graphics (A), Practical Research Training (A)

## **EXTRACURRICULAR EXPERIENCE**

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### **IIIS Soccer Team**

Jan 2024 – Feb 2025

*Captain*

*Beijing*

- Led a soccer team and managed the soccer club of IIIS.